

taller 10.

$$- f(x) = 0,15x^4 - 0,25x^2 + 3,5.$$

$$x = 0,6 ; h = 0,1 \Rightarrow x_{i+1} = 0,7 ; x_{i-1} = 0,5$$

Primera derivada.

hacia adelante

$$f'(x_i) \cong \frac{f(x_{i+1}) - f(x_i)}{h} = \frac{3,473515 - 3,42944}{0,1}$$

$$f'(0,6) \cong -0,23925$$

Hacia atrás:

$$f'(x_i) \cong \frac{f(x_i) - f(x_{i-1})}{h} = \frac{3,42944 - 3,446875}{0,1} =$$

$$f'(0,6) \cong -0,17435$$

Centrada:

$$f'(x_i) \cong \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} = \frac{3,473515 - 3,446875}{2(0,1)} =$$

$$f'(0,6) \cong -0,1668$$

Real

$$f'(0,6) = -0,1704$$

Segunda Derivada

$$x_{i+2} = 0,8 ; x_{i-2} = 0,4$$

hacia adelante:

$$f''(x_i) \approx \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} \Rightarrow$$

$$f''(x_i) \approx \frac{3,40744 - 2(3,47357) + 3,42944}{(0,1)^2} = \boxed{0,385}$$

hacia atrás:

$$f''(x_i) \approx \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2}))}{h^2} \Rightarrow$$

$$f''(0,6) \approx \frac{3,42944 - 2(3,446875) + 3,46384}{0,1^2} = \boxed{-0,047}$$

centrada:

$$f''(x_i) \approx \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2} \Rightarrow$$

$$f''(0,6) \approx \frac{3,47357 - 2(3,42944) + 3,446875}{0,1^2} = 0,757$$

Real

$$f''(0,6) = 0,748$$

b. $x_i = 0,6$; $h = 0,05$; $x_{i+1} = 0,65$; $x_{i-1} = 0,55$.

$$f(x_i) = 3,42944 ; f(x_{i+1}) = 3,42115 ; f(x_{i-1}) = 3,4381.$$

$$f'(x_i) = \boxed{-0,1704}$$

Primera Centralada:

$$f'(x) \approx \frac{f(x_{i+1}) - f(x_{i-1})}{2h} = \frac{3,42115 - 3,4381}{2(0,05)} = \boxed{-0,1695} \text{ Es mejor}$$

Segunda Centralada.

$$f''(x_i) = \boxed{0,148}$$

$$f''(x_i) \approx \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$$

$$f''(x_i) \approx \frac{(3,42114) - 2(3,42944) + (3,4381)}{(0,05)^2} \Rightarrow \boxed{0,144} \text{ Es mejor}$$